TestLink

Requirements Specification Document

Test Project: UniShare

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2026 Group UniShare

Table Of Contents

1. SRS-A-REG : UC-S2-001 Student Registration (detailed Use Case 1)

REQ-A-REG-01 : Public registration form and required fields

REQ-A-REG-02 : University and faculty selection

REQ-A-REG-03 : Minimum name length

REQ-A-REG-04 : Maximum name length of 100 (document-only gap)

REQ-A-REG-05 : Email syntax

REQ-A-REG-06 : Unique registration email and error recovery

REQ-A-REG-07 : Password minimum length

REQ-A-REG-08 : Password character classes

REQ-A-REG-09 : Password confirmation

REQ-A-REG-10 : Required uploaded ID and URL validation

REQ-A-REG-11 : Required academic selections

REQ-A-REG-12 : Faculty must belong to selected university (document-only gap)

REQ-A-REG-13 : Public upload accepts one image

REQ-A-REG-14 : Student ID file size maximum

REQ-A-REG-15 : Upload removal and recovery

REQ-A-REG-16 : Persist pending applicant with bcrypt and success UI

REQ-A-REG-17 : Welcome email is best effort

2. SRS-A-ADM : UC-S2-002 Admin Approval (detailed Use Case 2)

REQ-A-ADM-01 : Admin route and API authorization

REQ-A-ADM-02 : Pending list and safe user data

REQ-A-ADM-03 : Review identity details and full-size student ID

REQ-A-ADM-04 : Dashboard totals refresh after decisions

REQ-A-ADM-05 : Search/filter and sorting (document-only gap)

REQ-A-ADM-06 : Approval confirmation and valid transition

REQ-A-ADM-07 : Missing and nonpending targets rejected

REQ-A-ADM-08 : Approval email failure does not undo approval

REQ-A-ADM-09 : Rejection confirmation, optional reason and deletion

REQ-A-ADM-10 : Rejection email is best effort

3. SRS-A-LOG : UC-S2-003 Student Login (detailed Use Case 3)

REQ-A-LOG-01 : Valid credentials route by current role

REQ-A-LOG-02 : Generic invalid-credentials errors

REQ-A-LOG-03 : Required login fields

REQ-A-LOG-04 : Session persistence and authenticated-page redirects

REQ-A-LOG-05 : Fresh approval status without a new login

REQ-A-LOG-06 : Pending access restrictions and sign out

4. SRS-K-CRS : UC-S3-001 Create Course with Default Resources

REQ-CRS-01 : Course title validation

REQ-CRS-02 : Course description validation

REQ-CRS-03 : Course colour

REQ-CRS-04 : Four default resource cards

REQ-CRS-05 : Successful creation

REQ-CRS-06 : Only approved, logged-in students can create courses

5. SRS-K-MAN : Course management (list, open, edit, delete, favourite)

REQ-MAN-01 : Course list: owned and shared

REQ-MAN-02 : Course access control

REQ-MAN-03 : Edit course (owner only)

REQ-MAN-04 : Delete course (owner only)

REQ-MAN-05 : Favourite toggle

6. SRS-K-RES : Resource cards and files

REQ-RES-01 : Create custom resource card

REQ-RES-02 : Who may create or update resources

REQ-RES-03 : Delete resource (owner only)

REQ-RES-04 : Attach files to a card

REQ-RES-05 : Upload size, type and count limits

REQ-RES-06 : Remove a file

REQ-RES-07 : Deadline display

7. SRS-K-TT : UC-S4-005 Weekly Timetable with Events

REQ-TT-01 : Create timetable

REQ-TT-02 : Default timetable

REQ-TT-03 : Only favourite courses can be scheduled

REQ-TT-04 : Class event data

REQ-TT-05 : End time must be after start time

REQ-TT-06 : Time conflict detection

REQ-TT-07 : Maximum 3 classes per course per day

REQ-TT-08 : Edit and move a class

REQ-TT-09 : Delete a class

REQ-TT-10 : Timetable permissions

REQ-TT-11 : Invite students to a timetable

REQ-TT-12 : Accept, reject and leave a shared timetable

REQ-TT-13 : Class reminders

REQ-TT-14 : Update and delete timetable (owner only)

8. SRS-P-CHAT : UC-S3-011/012/013/019/020 Chat with AI Assistant

REQ-P-CHAT-01 : Only approved, logged-in students use the AI assistant

REQ-P-CHAT-02 : Message validation

REQ-P-CHAT-03 : New conversation is created on first message

REQ-P-CHAT-04 : Streaming response and persistence

REQ-P-CHAT-05 : Continue a previous conversation

REQ-P-CHAT-06 : Conversation history sidebar

REQ-P-CHAT-07 : Delete a conversation

REQ-P-CHAT-08 : Conversation privacy

REQ-P-CHAT-09 : AI unavailable handling

9. SRS-P-QUIZ : UC-S3-014/015 Generate and Attempt AI Quiz

REQ-P-QUIZ-01 : Quiz generation form and limits

REQ-P-QUIZ-02 : At least one question type

REQ-P-QUIZ-03 : Course / note context access

REQ-P-QUIZ-04 : Successful generation is saved

REQ-P-QUIZ-05 : Taking a quiz

REQ-P-QUIZ-06 : Automatic grading and score

REQ-P-QUIZ-07 : Quiz list and repeated attempts

REQ-P-QUIZ-08 : Quiz ownership and delete

10. SRS-P-PLAN : UC-S3-016/017 Generate and Track AI Study Plan

REQ-P-PLAN-01 : Study plan form and limits

REQ-P-PLAN-02 : Course access for study plans

REQ-P-PLAN-03 : Generated plan is saved with weeks and tasks

REQ-P-PLAN-04 : Track progress

REQ-P-PLAN-05 : Plan ownership and delete

REQ-P-PLAN-06 : Plan kept when the course is deleted

11. SRS-P-AIN : UC-S3-018 Enhance Notes with AI

REQ-P-AIN-01 : Enhancement request validation

REQ-P-AIN-02 : Only owner or contributor can enhance a note

REQ-P-AIN-03 : Enhancement result and history

12. SRS-P-NOTE : UC-S4-002 Real-Time Collaborative Notes

REQ-P-NOTE-01 : Open course notes

REQ-P-NOTE-02 : Only course members can open notes

REQ-P-NOTE-03 : Role-based editing (OWNER / CONTRIBUTOR / VIEWER)

REQ-P-NOTE-04 : Real-time synchronisation and persistence

REQ-P-NOTE-05 : Presence of live collaborators

REQ-P-NOTE-06 : Create pages and nested sub-pages

REQ-P-NOTE-07 : Delete pages and server-side permission checks

REQ-P-NOTE-08 : Network interruption

13. SRS-R-SHR : UC-S4-001 Course Sharing & Permissions

REQ-SHR-01 : Only the course owner can invite collaborators

REQ-SHR-02 : Invalid invitations are rejected

REQ-SHR-03 : Invitation carries a permission level

REQ-SHR-04 : Invitee accepts or declines a pending invitation

REQ-SHR-05 : VIEWER has read-only access

REQ-SHR-06 : CONTRIBUTOR can add but not delete

REQ-SHR-07 : Owner manages collaborators

REQ-SHR-08 : Faculty-restricted sharing

14. SRS-R-ART : UC-S4-003 Public Articles

REQ-ART-01 : Article input validation

REQ-ART-02 : New article starts as DRAFT

REQ-ART-03 : Unique URL slug

REQ-ART-04 : Publish workflow

REQ-ART-05 : Archive and delete own article

REQ-ART-06 : Edit own article

REQ-ART-07 : Public article listing

REQ-ART-08 : Article access control

REQ-ART-09 : View counter

REQ-ART-10 : My Articles

15. SRS-R-NOT : UC-S4-004 Notifications

REQ-NOT-01 : Event notifications

REQ-NOT-02 : Unread badge

REQ-NOT-03 : Recent notifications dropdown

REQ-NOT-04 : Mark one as read

REQ-NOT-05 : Mark all as read

REQ-NOT-06 : Notification privacy

REQ-NOT-07 : Notification preferences

REQ-NOT-08 : Reminder timing range

REQ-NOT-09 : Full notification history

1. Requirements Spec.: SRS-A-REG : UC-S2-001 Student Registration (detailed Use Case 1)

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (17/0)

Unauthenticated applicant supplies identity, academic details, password and student ID. Source: assignment/UniShare Iteration 2.md, detailed Use Case 1 and diagram UC-S2-001..007. Current UI is /signup (the document says /register), then /login after 3000 ms; src/app/(auth)/signup/page.tsx, src/server/api/routers/user.ts, src/app/api/uploadthing/core.ts, src/lib/email.ts, prisma/schema.prisma. REG-04 and REG-12 preserve document requirements absent from current server validation.

Requirement: REQ-A-REG-01 : Public registration form and required fields

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)

Source: Iteration 2 Use Case 1 steps 1-3, 23; src/app/(auth)/signup/page.tsx. /signup displays 'Join UNIShare', 'Full Name', 'University Email', 'Password', 'Confirm Password', 'University', 'Faculty', 'Student ID Photo' and 'Create Account'. The six text/select inputs are HTML required; missing input blocks form submission.

Related Test Cases	US-87 : Public registration displays all mandatory controls [Version :1]US-90 : Each empty required registration input blocks submission [Version :1]

Requirement: REQ-A-REG-02 : University and faculty selection

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)

Source: Iteration 2 Use Case 1 steps 6-13; user.ts getUniversities; signup/page.tsx. Universities and their faculties are sorted ascending by name. Faculty is disabled until a university is selected, contains only its faculties and resets to 'Select faculty' when university changes. Placeholders are 'Select university' and 'Select faculty'.

Related Test Cases	US-88 : University list is alphabetical and faculties are scoped [Version :1]US-89 : Changing university clears the previously selected faculty [Version :1]

Requirement: REQ-A-REG-03 : Minimum name length

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)

Source: Iteration 2 Use Case 1 step 23; user.ts register name schema. A name shorter than 2 characters is rejected with 'Name must be at least 2 characters'; 2 and 3 characters pass this constraint.

Related Test Cases	US-92 : Name minimum boundaries at 1, 2 and 3 characters [Version :1]

Requirement: REQ-A-REG-04 : Maximum name length of 100 (document-only gap)

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 step 23 specifies 2-100 characters; user.ts register and signup/page.tsx have no maximum. Business expectation: 99 and 100 accepted, 101 rejected without creating a user. Current code will accept 101; no maximum-length message exists. Record this as a document/code discrepancy when executed.	
Related Test Cases	US-93 : Document maximum name boundary at 99, 100 and 101 characters [Version :1]

Requirement: REQ-A-REG-05 : Email syntax	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 step 23; user.ts register email schema; signup/page.tsx. API rejects malformed/empty email with 'Invalid email address'. UI uses required type=email. No university-domain allowlist is implemented; syntactically valid example.com is accepted.	
Related Test Cases	US-94 : Email invalid partitions and valid nonuniversity domain [Version :1]

Requirement: REQ-A-REG-06 : Unique registration email and error recovery	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 steps 27-28, A1; user.ts register; prisma/schema.prisma User.email. An existing exact email is rejected with 'User with this email already exists'; no extra record is created. UI displays server error, clears its loading state and allows correction.	
Related Test Cases	US-95 : Existing approved and pending email cannot register twice [Version :1]US-105 : SMTP welcome failure preserves a successful registration [Version :1]US-106 : Duplicate-email server error can be corrected without reuploading [Version :1]

Requirement: REQ-A-REG-07 : Password minimum length	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 step 23; user.ts register; signup/page.tsx handleSubmit. Password minimum is 8 characters; failure text 'Password must be at least 8 characters'. There is no explicit application maximum; test 7, 8 and 9 characters with valid complexity.	
Related Test Cases	US-96 : Password length boundaries at 7, 8 and 9 characters [Version :1]

Requirement: REQ-A-REG-08 : Password character classes	
Version :	1

revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 secure-password description; user.ts register regex. New passwords require uppercase, lowercase and a digit. Missing any class returns 'Password must contain at least one uppercase letter, one lowercase letter, and one number'. Symbols are optional; seeded admin123 remains usable for login.	
Related Test Cases	US-97 : Password complexity decision table for all three classes [Version :1]

Requirement: REQ-A-REG-09 : Password confirmation	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 client validation; signup/page.tsx handleSubmit. Unequal password and confirmation show 'Passwords do not match' before the mutation; the form becomes editable again. Confirmation is a UI check, not an API field.	
Related Test Cases	US-98 : Mismatched confirmation blocks then corrected value succeeds [Version :1]

Requirement: REQ-A-REG-10 : Required uploaded ID and URL validation	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 steps 14-23; signup/page.tsx and user.ts register. Create Account is disabled without an uploaded URL. The handler guard says 'Please upload your student ID'. API studentIdUrl must be a URL or returns 'Invalid student ID URL'; the schema does not restrict the URL host.	
Related Test Cases	US-87 : Public registration displays all mandatory controls [Version :1]US-91 : Remove uploaded student ID then replace it [Version :1]US-99 : API rejects empty academic selections and invalid ID URLs [Version :1]

Requirement: REQ-A-REG-11 : Required academic selections	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 step 23; user.ts register universityId/facultyId. Empty universityId and facultyId return 'University is required' and 'Faculty is required'. These are nonempty-string checks; existing seeded IDs are used for valid cases.	
Related Test Cases	US-89 : Changing university clears the previously selected faculty [Version :1]US-90 : Each empty required registration input blocks submission [Version :1]US-99 : API rejects empty academic selections and invalid ID URLs [Version :1]

Requirement: REQ-A-REG-12 : Faculty must belong to selected university (document-only gap)	
Version :	1

revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 step 23 and A5; user.ts register; prisma/schema.prisma User/Faculty relations. Business expectation: reject a tampered request pairing UTM with a Universiti Malaya faculty, documented text 'Invalid faculty selection'. Current router only validates nonempty IDs and independent foreign keys, so does not enforce this relation.	
Related Test Cases	US-100 : Tampered faculty from a different university must be rejected [Version :1]

Requirement: REQ-A-REG-13 : Public upload accepts one image	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 steps 16-17, A4; api/uploadthing/core.ts studentIdUploader. Unauthenticated uploads accept image files, maximum file count 1; PDFs and a two-image batch are rejected. Completion returns uploadedBy='registration' and the file URL; UI shows 'Student ID uploaded'. Library-generated error wording is not fixed by app code.	
Related Test Cases	US-101 : Public image upload accepts JPG and PNG but rejects PDF [Version :1]US-103 : Two-file batch and failed upload do not enable registration [Version :1]

Requirement: REQ-A-REG-14 : Student ID file size maximum	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 step 17, A3; api/uploadthing/core.ts studentIdUploader. Image maxFileSize='4MB' (4194304 bytes); 4194303 and 4194304 are accepted, 4194305 rejected. The app formats uploader callback errors as 'Upload failed: <message>'.	
Related Test Cases	US-102 : ID image size boundaries at 4 MiB minus one, exact and plus one byte [Version :1]

Requirement: REQ-A-REG-15 : Upload removal and recovery	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 A2 and review step 21; signup/page.tsx upload callbacks. 'Remove' clears the selected URL, restores the uploader and disables Create Account. Upload failure displays 'Upload failed: <message>' and no successful ID state; retry can succeed.	
Related Test Cases	US-91 : Remove uploaded student ID then replace it [Version :1]US-103 : Two-file batch and failed upload do not enable registration [Version :1]

Requirement: REQ-A-REG-16 : Persist pending applicant with bcrypt and success UI	
Version :	1
revision:	1

Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 steps 29-32, 37-38; user.ts register; signup/page.tsx; prisma/schema.prisma. Successful registration stores role PENDING, null approvedAt, selected IDs, ID URL and bcrypt hash using 12 rounds. Response success=true with userId and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'. UI shows 'Registration Successful!', 'Your account has been created and is pending admin approval.' and 'Redirecting to login...', then /login after 3000 ms (document instead says /waiting-approval).	
Related Test Cases	US-104 : Successful applicant is PENDING with hashed password and welcome email [Version :1]US-105 : SMTP welcome failure preserves a successful registration [Version :1]US-106 : Duplicate-email server error can be corrected without reuploading [Version :1]US-131 : Deleted applicant cannot login and may register again for review [Version :1]

Requirement: REQ-A-REG-17 : Welcome email is best effort	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 1 steps 33-36, A6; user.ts register; src/lib/email.ts sendWelcomeEmail. Attempt welcome email to the applicant with subject 'Welcome to UNIShare - Registration Received'. SMTP failure logs 'Failed to send welcome email:' but registration still returns success and leaves the applicant PENDING; retry must not create a duplicate.	
Related Test Cases	US-104 : Successful applicant is PENDING with hashed password and welcome email [Version :1]US-105 : SMTP welcome failure preserves a successful registration [Version :1]

2. Requirements Spec.: SRS-A-ADM : UC-S2-002 Admin Approval (detailed Use Case 2)

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (10/0)

Admin reviews and approves/rejects applicants. Source: assignment/UniShare Iteration 2.md detailed Use Case 2 and diagram UC-S2-010..015; src/server/api/routers/admin.ts; src/app/(admin)/layout.tsx, admin/page.tsx, admin/approvals/page.tsx; src/lib/email.ts; prisma/schema.prisma. Search and interactive sorting are document-only gaps; bulk approval is future scope.

Requirement: REQ-A-ADM-01 : Admin route and API authorization	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 2 preconditions and steps 1-3; admin.ts adminProcedure; src/server/api/trpc.ts protectedProcedure; src/app/(admin)/layout.tsx; middleware.ts. Only ADMIN accesses admin procedures. Anonymous API requests return UNAUTHORIZED; authenticated APPROVED/PENDING users return FORBIDDEN 'Only administrators can perform this action'. Anonymous admin pages redirect /login; non-admins redirect through /courses (PENDING ends at /waiting-approval).	
Related Test Cases	US-109 : Admin login reaches approval dashboard [Version :1]US-115 : Anonymous and nonadmin users cannot access admin pages or procedures [Version :1]

Requirement: REQ-A-ADM-02 : Pending list and safe user data	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 2 steps 7-8; admin.ts getPendingUsers/getAllUsers. Pending list includes only role PENDING, newest createdAt first, with academic details and without password. getAllUsers also excludes password. Empty UI: 'All caught up!' and 'There are no pending approvals at the moment.'.	
Related Test Cases	US-116 : Pending list excludes approved/admin accounts, orders newest first and handles zero [Version :1]

Requirement: REQ-A-ADM-03 : Review identity details and full-size student ID	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 2 steps 10-15; admin/approvals/page.tsx. The current page uses cards with name, email, 'University:', 'Faculty:', 'Registered:', 'Student ID', 'View full size →', 'Approve' and 'Reject'. Clicking ID or full-size link opens the exact stored studentIdUrl in a new tab; no separate student-detail route is implemented.	
Related Test Cases	US-117 : Admin sees applicant identity and opens the stored student ID [Version :1]

Requirement: REQ-A-ADM-04 : Dashboard totals refresh after decisions

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 2 steps 5, 23-25; admin.ts getDashboardStats; approvals/page.tsx. Cards 'Pending Approvals', 'Approved Users', 'Total Users', 'Administrators' reflect role counts. Approve decreases pending and increases approved, leaving total/admin unchanged; reject decreases pending and total. Successful mutations invalidate pending list and statistics.	
Related Test Cases	US-118 : Dashboard role counts refresh after approve and reject [Version :1]

Requirement: REQ-A-ADM-05 : Search/filter and sorting (document-only gap)	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 2 step 8, diagram UC-S2-015; admin/approvals/page.tsx and admin.ts getPendingUsers. Business expectation: filter pending applicants by name/email and sort by date/name/university. Current cards expose no search/filter/sort controls; preserve as a gap rather than inventing labels or claiming these controls work.	
Related Test Cases	US-119 : Documented pending-applicant search and sorting are available [Version :1]

Requirement: REQ-A-ADM-06 : Approval confirmation and valid transition	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 2 steps 16-17; admin.ts approveUser; approvals/page.tsx. 'Are you sure you want to approve this user?' must be confirmed; cancel sends no mutation. Only PENDING transitions to APPROVED, setting approvedAt. Response success=true and '<name> has been approved successfully'; card disappears. While a decision is pending both action buttons are disabled and selected approval says 'Approving...'. No success toast is implemented.	
Related Test Cases	US-114 : Pending session observes admin approval without signing in again [Version :1]US-120 : Cancel approval leaves applicant pending [Version :1]US-121 : Confirm approval updates role, timestamp and email [Version :1]US-124 : Approval remains effective when approval email fails [Version :1]US-125 : Sequential replay of approval preserves first approval timestamp [Version :1]

Requirement: REQ-A-ADM-07 : Missing and nonpending targets rejected	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 2 A2/A3 and rejection alternative; admin.ts approveUser/rejectUser. Unknown userId returns NOT_FOUND 'User not found'. APPROVED/ADMIN targets return BAD_REQUEST 'User is not pending approval' for either operation; record and approvedAt stay unchanged. Sequential replay of approval is rejected; atomic concurrent safety is not claimed.	
Related Test Cases	US-122 : Approve refuses already approved and admin targets [Version :1]US-123 : Approve rejects nonexistent or empty target ID [Version :1]US-125 : Sequential replay of approval preserves first approval timestamp [Version :1]US-129 : Reject refuses approved/admin targets and missing users [Version :1]

Requirement: REQ-A-ADM-08 : Approval email failure does not undo approval	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 2 steps 18-21, A1; admin.ts approveUser; email.ts sendApprovalEmail. Attempt subject 'Your UNIShare Account Has Been Approved!' preceded by U+1F389 (party popper) and a space to applicant with dashboard URL NEXT_PUBLIC_APP_URL + '/courses'. Failure logs 'Failed to send approval email:' but response still succeeds and applicant remains APPROVED; the document's warning toast is not implemented.	
Related Test Cases	US-121 : Confirm approval updates role, timestamp and email [Version :1]US-124 : Approval remains effective when approval email fails [Version :1]

Requirement: REQ-A-ADM-09 : Rejection confirmation, optional reason and deletion	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 2 rejection alternative; admin.ts rejectUser; approvals/page.tsx; schema.prisma Role. UI prompts 'Enter rejection reason (optional):', then 'Are you sure you want to reject this user?'. Final cancel makes no change. Confirm deletes a PENDING applicant; there is no REJECTED enum or retain-account option. Response success=true and 'Registration for <name> has been rejected'. While pending, buttons disable and selected rejection says 'Rejecting...'. Deleted applicant cannot login and may register again.	
Related Test Cases	US-126 : Reject with reason sends explanation and deletes pending applicant [Version :1]US-127 : Optional rejection reason accepts blank and canceled prompt [Version :1]US-128 : Cancel final rejection confirmation preserves applicant [Version :1]US-130 : Rejection still deletes applicant when mail fails [Version :1]US-131 : Deleted applicant cannot login and may register again for review [Version :1]

Requirement: REQ-A-ADM-10 : Rejection email is best effort	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 2 rejection alternative and email-failure principle; admin.ts rejectUser; email.ts sendRejectionEmail. Attempt email before deletion, subject 'UNIShare Registration Update'; nonempty reason appears under 'Reason:', omitted/empty reason omits that section. SMTP failure logs 'Failed to send rejection email:' and deletion still completes.	
Related Test Cases	US-126 : Reject with reason sends explanation and deletes pending applicant [Version :1]US-127 : Optional rejection reason accepts blank and canceled prompt [Version :1]US-130 : Rejection still deletes applicant when mail fails [Version :1]

3. Requirements Spec.: SRS-A-LOG : UC-S2-003 Student Login (detailed Use Case 3)

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (6/0)

Credential authentication, role routing and sessions, including Admin login. Source: assignment/UniShare Iteration 2.md detailed Use Case 3 and diagram UC-S2-007..010; src/server/auth/config.ts; src/app/(auth)/login/page.tsx, waiting-approval/page.tsx; middleware.ts; src/app/(student)/layout.tsx; src/app/(admin)/layout.tsx. Implementation uses JWT sessions, not inserted Session rows or a Remember me control.

Requirement: REQ-A-LOG-01 : Valid credentials route by current role	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 3 steps 14-31; auth/config.ts authorize; login/page.tsx; middleware.ts; student/layout.tsx; admin/page.tsx. Password checked with bcrypt. APPROVED reaches /courses, PENDING /waiting-approval, ADMIN /admin/approvals via /courses. Successful UI shows 'Welcome Back!' and 'Taking you to your dashboard...'; PENDING page heading is 'Account Pending Approval'.	
Related Test Cases	US-107 : Approved student logs in with seeded credentials [Version :1]US-108 : Pending login cannot access student features and can sign out [Version :1]US-109 : Admin login reaches approval dashboard [Version :1]US-112 : Existing sessions bypass login and signup by role [Version :1]US-124 : Approval remains effective when approval email fails [Version :1]

Requirement: REQ-A-LOG-02 : Generic invalid-credentials errors	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 3 A1/A2; auth/config.ts authorize; login/page.tsx. Unknown email and wrong password show identical 'Invalid email or password', keep user on /login, and restore editable fields/'Sign In'. No authenticated session is issued.	
Related Test Cases	US-110 : Unknown email and wrong password have the same safe error [Version :1]US-131 : Deleted applicant cannot login and may register again for review [Version :1]

Requirement: REQ-A-LOG-03 : Required login fields	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 3 steps 7-8; login/page.tsx; auth/config.ts authorize. UI 'Email Address' is required type=email; 'Password' is required type=password. Missing values prevent browser submission. Direct authorize without credentials throws 'Missing email or password'; external UI errors are mapped to 'Invalid email or password'.	
Related Test Cases	US-111 : Missing and malformed login fields cannot authenticate [Version :1]

Requirement: REQ-A-LOG-04 : Session persistence and authenticated-page redirects	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 3 steps 21-27 and postconditions; auth/config.ts callbacks; middleware.ts; login/page.tsx; user.ts getCurrentSession. JWT-backed sessions persist across reload/new same-profile tabs, expose user id/role/universityId/facultyId without password, and existing-session /login or /signup returns to the role-appropriate destination.	
Related Test Cases	US-112 : Existing sessions bypass login and signup by role [Version :1]US-113 : JWT session survives reload and another tab without exposing password [Version :1]

Requirement: REQ-A-LOG-05 : Fresh approval status without a new login	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 3 role check/A3 and diagram UC-S2-009; auth/config.ts session callback; student/layout.tsx. Session reload reads current database role, so an Alice session issued while PENDING can reach /courses after admin approval without a new credentials submission.	
Related Test Cases	US-114 : Pending session observes admin approval without signing in again [Version :1]

Requirement: REQ-A-LOG-06 : Pending access restrictions and sign out	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Source: Iteration 2 Use Case 3 A3; waiting-approval/page.tsx; student/layout.tsx; middleware.ts; user.ts getCurrentSession. PENDING is routed away from student features to 'Account Pending Approval'. 'Sign Out' clears the session and returns /login; subsequent /courses redirects /login and getCurrentSession returns null.	
Related Test Cases	US-108 : Pending login cannot access student features and can sign out [Version :1]

4. Requirements Spec.: SRS-K-CRS : UC-S3-001 Create Course with Default Resources

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (6/0)

An approved student creates a course (title, description, colour); the system creates four default resource cards. Source: UniShare_Iteration_3.md Use Case 1 (UC-S3-001) and Sequence 1; src/server/api/routers/course.ts (create); src/components/courses/course-form.tsx; src/app/(student)/courses/new/page.tsx; middleware.ts; src/app/(student)/layout.tsx. Differences from the document: code has no course code / semester / icon fields; minimum title length is 1 (document: 2); an invalid colour is rejected, not defaulted; after creation the student is redirected to /courses (document: /courses/<id>).

Requirement: REQ-CRS-01 : Course title validation	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Title is mandatory, 1 to 100 characters (input has maxLength 100; API z.string().min(1,'Title is required').max(100)). The form trims the title; an empty or whitespace-only title is not submitted and shows the message 'Please enter a course title'.	
Related Test Cases	US-132 : Create a course with title, description and colour (main flow) [Version :1]US-133 : Title length boundaries (0, 1, 100, 101 characters) [Version :1]US-134 : Whitespace-only and padded title [Version :1]

Requirement: REQ-CRS-02 : Course description validation	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Description is optional, maximum 500 characters (textarea maxLength 500, API max(500)). An empty description is stored as no description.	
Related Test Cases	US-132 : Create a course with title, description and colour (main flow) [Version :1]US-135 : Description length boundaries (empty, 500, 501 characters) [Version :1]

Requirement: REQ-CRS-03 : Course colour	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Colour is mandatory and chosen from 8 swatches (Purple, Blue, Green, Orange, Red, Pink, Teal, Indigo); the first (Purple #a855f7) is preselected. The API accepts only '#' followed by exactly 6 hex digits (case-insensitive) and otherwise returns 'Invalid color format'.	
Related Test Cases	US-132 : Create a course with title, description and colour (main flow) [Version :1]US-136 : Colour selection - each swatch [Version :1]US-137 : Colour format validation in the API [Version :1]

Requirement: REQ-CRS-04 : Four default resource cards

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Creating a course automatically creates exactly 4 resource cards: Assignments (type ASSIGNMENT, files allowed), Tasks (TASK, no files), Content (CONTENT, files allowed), Notes (NOTES, no files).	
Related Test Cases	US-138 : Four default resource cards are created [Version :1]

Requirement: REQ-CRS-05 : Successful creation	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
After a valid submission the system shows 'Course created successfully!', redirects to /courses and the course appears under 'My Courses' with the creator as owner (role OWNER, createdBy = current user). Duplicate titles are allowed.	
Related Test Cases	US-132 : Create a course with title, description and colour (main flow) [Version :1]US-138 : Four default resource cards are created [Version :1]US-139 : Duplicate course titles are allowed [Version :1]

Requirement: REQ-CRS-06 : Only approved, logged-in students can create courses	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Visitors who are not logged in are redirected to /login; users with role PENDING are redirected to /waiting-approval and cannot use /courses/new. The API requires a session (UNAUTHORIZED otherwise). Iteration 3 alternative flow A3 additionally requires a 403 'Your account is pending approval' from the server.	
Related Test Cases	US-140 : Course creation blocked for visitors and PENDING students [Version :1]US-141 : course.create called without a session or with a PENDING session [Version :1]

5. Requirements Spec.: SRS-K-MAN : Course management (list, open, edit, delete, favourite)

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (5/0)

Owner and collaborator handling of an existing course. Source: UniShare_Iteration_3.md Use Case 1 postconditions and Data Dictionary (Course, Favorite); src/server/api/routers/course.ts (getAll, getUserCourses, getById, update, delete, toggleFavorite); src/app/(student)/courses/page.tsx, courses/[id]/page.tsx, courses/[id]/settings/page.tsx; src/components/courses/*.

Requirement: REQ-MAN-01 : Course list: owned and shared	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
/courses lists the student's own courses under 'My Courses' and courses where the student is an ACCEPTED collaborator under 'Shared With Me'. PENDING or REJECTED invitations and other students' courses are not listed. A student without courses sees 'No courses yet'.	
Related Test Cases	US-142 : Course list shows owned and shared courses only [Version :1]

Requirement: REQ-MAN-02 : Course access control	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A course can be opened only by its owner or an ACCEPTED collaborator; the page shows the role badge OWNER / CONTRIBUTOR / VIEWER. Others get 'You don't have access to this course'; an unknown id gives 'Course not found' (page 'Course Not Found').	
Related Test Cases	US-143 : Course access by role [Version :1]

Requirement: REQ-MAN-03 : Edit course (owner only)	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The owner can change title, description and colour in Course Settings > General (same validation as creation) and sees 'Course updated successfully!'. Any other user gets 'Only the course owner can update course details'; the settings page shows 'Access Denied - Only the course owner can access settings.' to non-owners. Clearing the description must remove it.	
Related Test Cases	US-144 : Owner edits title, description and colour [Version :1]US-145 : Non-owner cannot edit or open settings [Version :1]US-146 : Clearing the description when editing a course [Version :1]

Requirement: REQ-MAN-04 : Delete course (owner only)	
Version :	1
revision:	1

Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Course Settings > Danger Zone asks 'Are you sure you want to delete "<title>"? This action cannot be undone and will delete all resources and collaborator access.'. OK deletes the course with its resources, collaborators and favourites and shows 'Course deleted successfully'; Cancel changes nothing. Non-owners get 'Only the course owner can delete the course'.	
Related Test Cases	US-147 : Delete course - confirm and cancel [Version :1]US-148 : Non-owner cannot delete a course [Version :1]

Requirement: REQ-MAN-05 : Favourite toggle	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The star on the course page toggles the favourite flag of the current student only (unique per student and course): 'Added to favorites' / 'Removed from favorites'. Favourites feed the timetable course list.	
Related Test Cases	US-149 : Toggle favourite on and off [Version :1]

6. Requirements Spec.: SRS-K-RES : Resource cards and files

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (7/0)

Custom resource cards, deadlines and file attachments inside a course. Source: UniShare_Iteration_3.md Use Case 1 step 26 + Data Dictionary (ResourceCard, Resource); src/server/api/routers/resource.ts; src/components/resources/resource-form.tsx, resource-card.tsx, file-upload-modal.tsx; src/app/api/uploadthing/core.ts (courseResourceUploader); src/app/(student)/courses/[id]/page.tsx.

Requirement: REQ-RES-01 : Create custom resource card	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
'Add Custom Resource' opens 'Create Custom Resource'. Title is mandatory 1-100 characters (empty shows 'Please enter a title'), description optional max 500, 'Allow File Uploads' toggle (default ON), optional deadline ('Set a Deadline'). The card is stored with type CUSTOM and 'Resource created successfully!' is shown.	
Related Test Cases	US-150 : Owner creates a custom resource card [Version :1]US-151 : Resource title and description boundaries [Version :1]US-152 : Deadline states: future, within 3 days, overdue [Version :1]

Requirement: REQ-RES-02 : Who may create or update resources	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the course owner and ACCEPTED CONTRIBUTORs can create/update resources and see 'Add Custom Resource'. VIEWERs and outsiders get 'Only course owner and contributors can create resources' / 'Only course owner and contributors can update resources'; unknown course/resource gives 'Course not found' / 'Resource not found'.	
Related Test Cases	US-150 : Owner creates a custom resource card [Version :1]US-153 : Create/update permission matrix [Version :1]

Requirement: REQ-RES-03 : Delete resource (owner only)	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the owner can delete resources: the API returns 'Only the course owner can delete resources' to everybody else. In the UI the delete button exists only for CUSTOM cards, asks 'Are you sure you want to delete this resource?' and shows 'Resource deleted successfully'. The 4 default cards have no delete button.	
Related Test Cases	US-154 : Delete resource - owner only, custom cards only in the UI [Version :1]

Requirement: REQ-RES-04 : Attach files to a card

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Files can be added (button 'Add Files' / 'Add More Files') only to cards with allowFiles = true and only by the owner or a CONTRIBUTOR. Otherwise the API returns 'This resource does not allow file uploads' or 'You don't have permission to add files to this resource'. File URLs must be valid URLs. Success shows 'Files added successfully'. Cards without files show 'No files uploaded yet'.	
Related Test Cases	US-155 : Attach files to a card that allows uploads [Version :1]US-156 : File upload refused for cards without uploads and invalid URLs [Version :1]

Requirement: REQ-RES-05 : Upload size, type and count limits	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
courseResourceUploader accepts images, PDF and text up to 16MB (max 10 files), Word/Excel/PowerPoint up to 16MB (max 10 files), video up to 64MB (max 5 files), ZIP up to 32MB (max 5 files); anything else, any larger file or more files is rejected ('Failed to upload files: ...'). Upload requires a logged-in user ('Unauthorized').	
Related Test Cases	US-157 : Upload limits: size, type and count [Version :1]

Requirement: REQ-RES-06 : Remove a file	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Owner and CONTRIBUTOR can remove a file URL from a card; others get 'You don't have permission to remove files from this resource'. Only the chosen file is removed.	
Related Test Cases	US-158 : Remove a file from a card [Version :1]

Requirement: REQ-RES-07 : Deadline display	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A card with a deadline shows 'Due: <MMM dd, yyyy 'at' h:mm a>'. Within the next 3 days it additionally shows the badge 'Soon' (amber); after the deadline it shows 'Overdue: <date>' (red) and no 'Soon' badge.	
Related Test Cases	US-152 : Deadline states: future, within 3 days, overdue [Version :1]

7. Requirements Spec.: SRS-K-TT : UC-S4-005 Weekly Timetable with Events

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (14/0)

Student creates timetables, adds weekly class events for favourite courses, detects conflicts, shares the timetable and receives class reminders. Source: UniShare_Iteration_4.md Use Case 5 (UC-S4-005); src/server/api/routers/timetable.ts; src/components/timetable/*; src/app/(student)/timetable/page.tsx; src/app/api/cron/class-reminders/route.ts. Differences from the document: events are weekly recurring by day-of-week (no recurrence rule JSON, end date or multiple days per event); the conflict message is 'Time conflict detected / Overlaps with <title> (start - end)'; reminders use the user's reminder timing (5-30 min, default 10) not a fixed 1 hour; the check 'start before end' and the conflict check are done only in the browser, not in the API.

Requirement: REQ-TT-01 : Create timetable	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
'Create Timetable' (or 'Create Your First Timetable' when the student has none: 'No Timetables Yet') needs a Name of 1-100 characters (API message 'Name is required') and an optional Description of max 500. The new timetable is listed under 'My Timetables' and the creator is the owner.	
Related Test Cases	US-159 : Create a timetable (first-time and name boundaries) [Version :1]

Requirement: REQ-TT-02 : Default timetable	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Timetable Settings > 'Set as Default' makes the timetable the student's default (button changes to 'Currently Default', badge 'Default'); /timetable opens the default timetable first. Only owner or accepted collaborator may set it ('You don't have access to this timetable'). Deleting or leaving the timetable clears the default.	
Related Test Cases	US-160 : Set, keep and lose the default timetable [Version :1]US-173 : Timetable invitation lifecycle: accept, reject, leave [Version :1]US-175 : Rename and delete a timetable (owner only) [Version :1]

Requirement: REQ-TT-03 : Only favourite courses can be scheduled	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
'Add Class' lists only the student's favourite courses ('Only your favorited courses are shown'). Without favourites the modal shows 'No Favorite Courses - You need to favorite courses before adding them to your timetable'. The API returns 'You can only add favorited courses to your timetable' for other courses.	
Related Test Cases	US-161 : Add a weekly class to the timetable (main flow) [Version :1]US-162 : Add Class with no favourite courses [Version :1]US-163 : API refuses classes for non-favourite courses [Version :1]

Requirement: REQ-TT-04 : Class event data	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A class has Course (required), Title (required, 1-100), Day of Week (Sunday=0 ... Saturday=6), Start Time and End Time (HH:MM) and optional Location (max 100). It is stored as weekly recurring and appears in the week view on that weekday in the course colour. API: dayOfWeek 0-6, times must match ^\d{2}:\d{2}$ ('Invalid time format (HH:MM)').	
Related Test Cases	US-161 : Add a weekly class to the timetable (main flow) [Version :1]US-164 : Class title and location length boundaries [Version :1]

Requirement: REQ-TT-05 : End time must be after start time	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
If End Time is not later than Start Time the form shows 'End time must be after start time' and 'Add Class' is disabled; the equal-times case is also invalid. Dragging/resizing to an invalid range shows 'Invalid time range'. The API must also reject such events.	
Related Test Cases	US-165 : End time must be after start time [Version :1]US-168 : Edit a class in the details dialog [Version :1]

Requirement: REQ-TT-06 : Time conflict detection	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Two classes on the same weekday must not overlap (A.start < B.end and B.start < A.end). The form shows 'Time conflict detected' and 'Overlaps with <title> (<start> - <end>)' and disables 'Add Class'. Back-to-back classes (end = next start) and the same time on another weekday are allowed.	
Related Test Cases	US-166 : Time conflict detection boundaries [Version :1]US-167 : Maximum 3 classes per course per day [Version :1]US-168 : Edit a class in the details dialog [Version :1]

Requirement: REQ-TT-07 : Maximum 3 classes per course per day	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A course can have at most 3 classes on one weekday. With 3 existing the form shows 'Maximum 3 classes per course per day. This course already has 3 classes on this day.' and blocks submit; with 1 or 2 it shows 'This course has N class(es) on this day. You can add M more.'.	
Related Test Cases	US-167 : Maximum 3 classes per course per day [Version :1]

Requirement: REQ-TT-08 : Edit and move a class	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Owner and CONTRIBUTORs can edit a class (Class Details > Edit Class > 'Save Changes') or drag/resize it in the calendar. The same time-range and conflict rules apply, ignoring the class itself. Conflict while dragging shows 'Cannot move event: Time conflict with another class' / 'Cannot resize event: Time conflict with another class'.	
Related Test Cases	US-168 : Edit a class in the details dialog [Version :1]US-169 : Drag and resize a class in the calendar [Version :1]US-171 : Timetable role matrix (owner, contributor, viewer, outsider) [Version :1]

Requirement: REQ-TT-09 : Delete a class	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Owner and CONTRIBUTORs can delete a class: 'Delete this class?' with 'Yes, Delete' / 'Cancel'. Yes removes the class from every week; Cancel keeps it.	
Related Test Cases	US-170 : Delete a class [Version :1]US-171 : Timetable role matrix (owner, contributor, viewer, outsider) [Version :1]

Requirement: REQ-TT-10 : Timetable permissions	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Owner: everything. CONTRIBUTOR: add/edit/delete classes. VIEWER: read only, no 'Add Class'. Users without an accepted role cannot open the timetable ('You don't have access to this timetable'). API messages: 'You don't have permission to add events to this timetable' / '... update this event' / '... delete this event'.	
Related Test Cases	US-171 : Timetable role matrix (owner, contributor, viewer, outsider) [Version :1]

Requirement: REQ-TT-11 : Invite students to a timetable	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the owner can invite ('Only the owner can invite collaborators') through Share > 'Search for students': the query needs at least 3 characters ('Search query must be at least 3 characters'), only APPROVED users are found (max 10), the owner and already invited users are excluded, role Viewer or Contributor. Inviting yourself ('You can't invite yourself') or an invited user ('User already invited') is rejected. A TIMETABLE_INVITATION notification 'New timetable invitation' is sent.	
Related Test Cases	US-172 : Invite a student to a timetable (search length, role, duplicates) [Version :1]

Requirement: REQ-TT-12 : Accept, reject and leave a shared timetable	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A PENDING invitation appears under 'Timetable Invitations' with Accept / Reject. Accept -> ACCEPTED and the timetable shows under 'Shared with Me'; Reject -> REJECTED. The owner is notified ('Invitation accepted' / 'Invitation declined'). A collaborator can 'Leave Timetable' (Confirm Leave); the owner cannot ('You cannot leave a timetable you own. Delete it instead.').	
Related Test Cases	US-173 : Timetable invitation lifecycle: accept, reject, leave [Version :1]

Requirement: REQ-TT-13 : Class reminders	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The cron endpoint /api/cron/class-reminders (Authorization: Bearer CRON_SECRET, else 401) creates a CLASS_REMINDER notification 'Class starting in <N> minutes' / '<course> at <HH:MM>' for owners and accepted collaborators when the class starts in N minutes (+-1), N = the user's reminder timing (default 10). At most one reminder per class and day.	
Related Test Cases	US-174 : Class reminder is created at the configured time [Version :1]

Requirement: REQ-TT-14 : Update and delete timetable (owner only)	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the owner can rename/describe ('Only the owner can update this timetable') and delete ('Only the owner can delete this timetable') a timetable. Deleting removes its classes and collaborators and clears the default for every user who had it.	
Related Test Cases	US-175 : Rename and delete a timetable (owner only) [Version :1]

8. Requirements Spec.: SRS-P-CHAT : UC-S3-011/012/013/019/020 Chat with AI Assistant

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (9/0)

Student chats with the AI tutor, keeps a conversation history, continues and deletes conversations. Source: UniShare_Iteration_3.md AI Learning Assistant Use Case, Sequence 2, Use Case 2 (UC-S3-012); src/app/(student)/ai/[[...id]]/page.tsx, src/components/ai/ai-chat-interface.tsx, conversation-history-sidebar.tsx, src/app/api/chat/route.ts, src/server/api/routers/ai.ts.

Requirement: REQ-P-CHAT-01 : Only approved, logged-in students use the AI assistant	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The /ai page shall be available only to authenticated users with role APPROVED; a PENDING user is redirected to /waiting-approval. POST /api/chat without a session returns 401 'Unauthorized'.	
Related Test Cases	US-46 : Chat access without approval, session or conversation id [Version :1]

Requirement: REQ-P-CHAT-02 : Message validation	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A message shall not be empty (whitespace only counts as empty; the Send button stays disabled and nothing is sent) and shall be at most 5000 characters ('Message must be under 5000 characters', Iteration 3 A2). Implementation note: tRPC ai.sendMessage accepts 1-10000 characters.	
Related Test Cases	US-43 : Empty and whitespace-only message is not sent [Version :1]US-44 : Message length boundaries (4999, 5000, 5001 characters) [Version :1]

Requirement: REQ-P-CHAT-03 : New conversation is created on first message	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Sending the first message in a new chat creates an AiConversation owned by the user with the client-generated id, temperature 0-2 (default 0.7) and an AI-generated title of at most 6 words (fallback: first 50 characters of the message). The URL changes to /ai/<id> and the conversation appears at the top of the sidebar. A request without conversationId returns 400 'No conversation ID provided'.	
Related Test Cases	US-42 : Start a new AI conversation (main flow) [Version :1]US-46 : Chat access without approval, session or conversation id [Version :1]US-49 : New Chat and untitled conversation [Version :1]

Requirement: REQ-P-CHAT-04 : Streaming response and persistence	
Version :	1

revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The AI answer (model badge 'Gemini 2.5 Flash') streams into the chat while a loader is shown; input submission is disabled while status is submitted/streaming. When the stream finishes the user and assistant messages are saved (assistant message with tokensUsed) and the conversation updatedAt is set.	
Related Test Cases	US-42 : Start a new AI conversation (main flow) [Version :1]US-45 : Continue a previous conversation with context [Version :1]

Requirement: REQ-P-CHAT-05 : Continue a previous conversation	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Opening /ai/<id> loads all messages of that conversation oldest first; a follow-up message is answered with the earlier messages as context and is appended to the same conversation.	
Related Test Cases	US-45 : Continue a previous conversation with context [Version :1]

Requirement: REQ-P-CHAT-06 : Conversation history sidebar	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The sidebar lists only the user's conversations, most recently updated first, 5 at a time; 'Load More' adds 10 more and the footer shows 'Showing <n> of <total>'. A conversation without title shows 'Untitled Chat'. 'New Chat' opens an empty chat at /ai. API page size 1-100.	
Related Test Cases	US-48 : Sidebar paging boundaries (5, 6 and 16 conversations) [Version :1]US-49 : New Chat and untitled conversation [Version :1]

Requirement: REQ-P-CHAT-07 : Delete a conversation	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The delete icon asks 'Delete this conversation? This action cannot be undone.'; on OK the conversation and all its messages are deleted (cascade) and it disappears from the sidebar; on Cancel nothing changes.	
Related Test Cases	US-50 : Delete conversation - confirm and cancel [Version :1]

Requirement: REQ-P-CHAT-08 : Conversation privacy	
Version :	1
revision:	1
Author:	admin

Status	Valid
Type	Use Case
Coverage	100% (1/1)
A user can read, continue or delete only his own conversations. Another user's conversation id gives NOT_FOUND 'Conversation not found' (Iteration 3 A3: no access to other users' conversations).	
Related Test Cases	US-51 : User cannot read, continue or delete another user's conversation [Version :1]

Requirement: REQ-P-CHAT-09 : AI unavailable handling	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
If the AI server cannot be reached the chat shows 'Unable to connect to AI server. Please make sure Ollama is running.'; a server-side failure returns 500 'Failed to process AI request'. The user message is kept and the student can retry.	
Related Test Cases	US-47 : AI server unavailable [Version :1]

9. Requirements Spec.: SRS-P-QUIZ : UC-S3-014/015 Generate and Attempt AI Quiz

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (8/0)

Student generates a practice quiz with AI for a course and attempts it with automatic grading. Source: UniShare_Iteration_3.md Sequence 3, Use Case 3 (UC-S3-014), UC-S3-015; src/server/api/routers/ai.ts (generateQuiz, getQuizzes, getQuiz, submitQuizAttempt, deleteQuiz); src/server/ai/quiz-generator.ts; src/components/ai/quiz-generator-form.tsx, quiz-taker.tsx, quiz-results.tsx, quiz-list.tsx; course page tab 'Quizzes'.

Requirement: REQ-P-QUIZ-01 : Quiz generation form and limits	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Course > Quizzes > Generate Quiz shows Topic * (required, trimmed; empty gives 'Please enter a topic'), Number of Questions slider 5-50 in steps of 5 (default 10), Difficulty Easy/Medium/Hard (default Medium) and Question Types Multiple Choice, True/False (checked by default) and Short Answer. The API accepts questionCount 1-50 and difficulty easy|medium|hard.	
Related Test Cases	US-52 : Generate a quiz for own course (main flow) [Version :1]US-53 : Topic is required [Version :1]US-55 : Question count boundaries [Version :1]US-56 : Difficulty and question type partitions [Version :1]

Requirement: REQ-P-QUIZ-02 : At least one question type	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Submitting with no question type checked shows 'Please select at least one question type' and no quiz is generated.	
Related Test Cases	US-54 : No question type selected [Version :1]

Requirement: REQ-P-QUIZ-03 : Course / note context access	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A quiz may be generated for a course or note only by the course owner or an ACCEPTED collaborator. Unknown course: 'Course not found'; no access: 'You don't have access to this course'. Unknown note: 'Note not found'; no access: 'You don't have access to this note'.	
Related Test Cases	US-57 : Quiz generation access by course relation [Version :1]

Requirement: REQ-P-QUIZ-04 : Successful generation is saved	
Version :	1
revision:	1

Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
On success the quiz and its questions (in generated order) are stored for the user with the topic as prompt and token usage, toast 'Quiz generated successfully!' is shown and the quiz opens for taking. While generating, the form is disabled and the button shows 'Generating...'. If the AI answer is not valid quiz JSON the error 'Failed to generate quiz. The AI response was not in the expected format.' is shown and nothing is saved.	
Related Test Cases	US-52 : Generate a quiz for own course (main flow) [Version :1]US-58 : Invalid AI response is not saved [Version :1]

Requirement: REQ-P-QUIZ-05 : Taking a quiz	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The quiz taker shows '<answered> of <total> answered', '<p>% complete', one button per question, Previous (disabled on question 1), Next, and on the last question 'Submit Quiz', which stays disabled until every question is answered ('Please answer all questions before submitting'). Multiple choice shows the generated options, True/False shows True and False, Short Answer shows 'Type your answer here...'.	
Related Test Cases	US-56 : Difficulty and question type partitions [Version :1]US-59 : Submit is enabled only when all questions are answered [Version :1]

Requirement: REQ-P-QUIZ-06 : Automatic grading and score	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Multiple choice and True/False answers are correct only on exact match with the stored correct answer; Short Answer is compared trimmed and case-insensitive. Score = correct / total questions x 100. The attempt is saved with completedAt. Results show the score with message: >=90 'Excellent!', >=80 'Great job!', >=70 'Good work!', >=60 'Not bad!', otherwise 'Keep studying!'.	
Related Test Cases	US-60 : Score bands on the results page [Version :1]US-61 : Answer matching rules per question type [Version :1]US-62 : Attempt is saved, best score and retake [Version :1]

Requirement: REQ-P-QUIZ-07 : Quiz list and repeated attempts	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The Quizzes tab lists the user's quizzes for the course newest first with 'Best Score: <n>%'; with none it shows 'No quizzes yet'. A quiz can be attempted any number of times ('Retake Quiz'); every attempt is kept.	
Related Test Cases	US-62 : Attempt is saved, best score and retake [Version :1]US-64 : Delete own quiz [Version :1]

Requirement: REQ-P-QUIZ-08 : Quiz ownership and delete	
Version :	1

revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the creator can open, attempt or delete a quiz; otherwise 'Quiz not found'. Delete asks 'Are you sure you want to delete this quiz? This action cannot be undone.' and removes the quiz with its questions and attempts.	
Related Test Cases	US-63 : Another user's quiz cannot be opened, attempted or deleted [Version :1]US-64 : Delete own quiz [Version :1]

10. Requirements Spec.: SRS-P-PLAN : UC-S3-016/017 Generate and Track AI Study Plan

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (6/0)

Student generates a weekly study plan for a course and tracks task completion. Source: UniShare_Iteration_3.md Sequence 4, UC-S3-016, UC-S3-017, Data Description 'Study Plan Generation'; src/server/api/routers/ai.ts (generateStudyPlan, getStudyPlans, updateStudyPlanTask, deleteStudyPlan); src/server/ai/study-plan-generator.ts; src/components/ai/study-plan-generator-form.tsx, study-plan-viewer.tsx, study-plan-list.tsx; prisma/schema.prisma (AiStudyPlan).

Requirement: REQ-P-PLAN-01 : Study plan form and limits	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Generate Study Plan requires a course ('Please select a course'); Number of Weeks 1-16 (default 4; Iteration 3 states 1-12), Hours Per Week 1-40 (default 5), Learning Goal (Optional) and Target Deadline (Optional, dates before today cannot be picked). Values outside the ranges are rejected by the API.	
Related Test Cases	US-65 : Generate a study plan (main flow) [Version :1]US-66 : Weeks and hours boundaries [Version :1]US-67 : Course is required and deadline cannot be in the past [Version :1]

Requirement: REQ-P-PLAN-02 : Course access for study plans	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the course owner or an ACCEPTED collaborator can generate a plan for a course. Unknown course: 'Course not found'; no access: 'You don't have access to this course'.	
Related Test Cases	US-68 : Study plan access by course relation [Version :1]

Requirement: REQ-P-PLAN-03 : Generated plan is saved with weeks and tasks	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
On success the plan is saved with its weeks (ordered by week number) and tasks (in order), toast 'Study plan generated successfully!' is shown. Without goal the stored prompt is 'Study plan for course'. With a deadline endDate = deadline and startDate = deadline - weeks x 7 days. Invalid AI JSON gives 'Failed to generate study plan. The AI response was not in the expected format.'	
Related Test Cases	US-65 : Generate a study plan (main flow) [Version :1]US-67 : Course is required and deadline cannot be in the past [Version :1]

Requirement: REQ-P-PLAN-04 : Track progress

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Ticking a task marks it completed (completedAt set, toast 'Task updated'); unticking clears completedAt. Overall Progress = completed tasks / all tasks x 100 (rounded, 0% when there are no tasks) and every week shows its Week Progress; a week at 100% is shown as complete.	
Related Test Cases	US-69 : Track task completion and progress [Version :1]

Requirement: REQ-P-PLAN-05 : Plan ownership and delete	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the owner of a plan can update its tasks ('Task not found' otherwise) or delete it ('Study plan not found'). Delete asks 'Are you sure you want to delete this study plan? This action cannot be undone.'. With no plans the tab shows 'No study plans yet'.	
Related Test Cases	US-70 : Plan ownership and delete [Version :1]

Requirement: REQ-P-PLAN-06 : Plan kept when the course is deleted	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Study plans are linked to a course by an optional courseId with SET NULL on course delete, so plans are preserved (Iteration 3 Data Description).	
Related Test Cases	US-71 : Study plan survives course deletion [Version :1]

11. Requirements Spec.: SRS-P-AIN : UC-S3-018 Enhance Notes with AI

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (3/0)

Student uses AI to generate, improve, summarize or expand a course note. Source: UniShare_Iteration_3.md UC-S3-018, Data Description 'Note Enhancement'; src/app/api/ai/generate-note/route.ts; src/server/api/routers/ai.ts (generateNote, getNoteGenerations).

Requirement: REQ-P-AIN-01 : Enhancement request validation	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Type must be one of GENERATE, IMPROVE, SUMMARIZE, EXPAND and prompt must be 1-5000 characters; otherwise 400 'Invalid request'. Temperature, if given, 0-2.	
Related Test Cases	US-72 : Each enhancement type returns content and is recorded [Version :1]US-73 : Invalid enhancement requests [Version :1]

Requirement: REQ-P-AIN-02 : Only owner or contributor can enhance a note	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The course owner and ACCEPTED CONTRIBUTORs can enhance a note. A VIEWER or unrelated user gets 403 'Forbidden: You don't have permission to edit this note'; unknown note 404 'Note not found'; no session 401 'Unauthorized'.	
Related Test Cases	US-74 : Enhancement permission by role [Version :1]

Requirement: REQ-P-AIN-03 : Enhancement result and history	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A successful call returns success, generatedText (markdown) and tokensUsed and stores an AiGeneratedNote with prompt, type, content before and after. getNoteGenerations returns only the caller's generations for that note, newest first (page size 1-50, default 20). AI provider errors return 503.	
Related Test Cases	US-72 : Each enhancement type returns content and is recorded [Version :1]US-74 : Enhancement permission by role [Version :1]

12. Requirements Spec.: SRS-P-NOTE : UC-S4-002 Real-Time Collaborative Notes

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (8/0)

Course owner and collaborators edit nested note pages together in real time (Liveblocks + BlockNote). Source: UniShare_Iteration_4.md Use Case Diagram 2 and Use Case 2 (UC-S4-002), Data Description 'Note Table'; src/app/(student)/courses/[id]/notes/[[...pageId]]/page.tsx, src/components/notes/*, src/components/resources/resource-card.tsx, src/app/api/notes/create, src/app/api/notes/[id], src/app/api/liveblocks-auth/route.ts.

Requirement: REQ-P-NOTE-01 : Open course notes	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
'Open Collaborative Notes' on the Notes card opens /courses/<id>/notes. If the course has no pages and the user can edit, a page 'Getting Started' is created and opened; a VIEWER sees 'No pages have been created yet. Ask a contributor or owner to create one.'. Without page id, or with an unknown / other course page id, the first page is opened.	
Related Test Cases	US-75 : Owner opens notes for the first time [Version :1]US-76 : Viewer opens a course without pages [Version :1]US-80 : Unknown or other course page id [Version :1]

Requirement: REQ-P-NOTE-02 : Only course members can open notes	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the owner and ACCEPTED collaborators can open the notes: others are redirected to /courses/<id>, logged-out users to /login, unknown course to /courses. Liveblocks auth: no session 401 'Unauthorized', missing room 400 'Bad Request - Room ID missing', unknown page 404 'Page not found', non-member 403 'Forbidden'.	
Related Test Cases	US-77 : Notes access by user relation [Version :1]US-79 : Liveblocks authentication endpoint [Version :1]

Requirement: REQ-P-NOTE-03 : Role-based editing (OWNER / CONTRIBUTOR / VIEWER)	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The header shows the role badge OWNER, CONTRIBUTOR or VIEWER. OWNER and CONTRIBUTOR can edit content, titles and pages. VIEWER sees 'Viewing in Read-Only Mode', cannot type in the editor or title and sees no add/delete page controls; the server must not accept a VIEWER's edits (Iteration 4: VIEWER read-only).	
Related Test Cases	US-75 : Owner opens notes for the first time [Version :1]US-78 : Viewer is read-only (UI and server) [Version :1]

Requirement: REQ-P-NOTE-04 : Real-time synchronisation and persistence	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Text typed by one member appears for the other members of the same page in real time; concurrent edits, also of the same block, merge without conflict and all editors converge (Yjs CRDT). Content is persisted by Liveblocks ('All changes saved') and is still there after reload.	
Related Test Cases	US-81 : Two students edit the same page in real time [Version :1]US-82 : Simultaneous edit of the same text converges [Version :1]US-86 : Offline edits sync after reconnect [Version :1]

Requirement: REQ-P-NOTE-05 : Presence of live collaborators	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The 'Active Collaborators' bar shows the current user ('<name> (You)') and up to 5 other online users as coloured initials, '+<n>' for more, and '<count> online'. When a user leaves, his avatar disappears.	
Related Test Cases	US-81 : Two students edit the same page in real time [Version :1]US-83 : Presence list boundaries and leaving [Version :1]

Requirement: REQ-P-NOTE-06 : Create pages and nested sub-pages	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The '+' button in 'Pages' creates an 'Untitled' page placed after the last page and opens it; 'Add sub-page' creates a child page shown indented under its parent (expand/collapse). Each page has its own room. The title is edited inline and saved on Enter or leaving the field.	
Related Test Cases	US-84 : Create page and nested sub-page [Version :1]

Requirement: REQ-P-NOTE-07 : Delete pages and server-side permission checks	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Delete asks 'Are you sure you want to delete this page?'; deleting a page also deletes its sub-pages; if the open page is deleted the first remaining page opens. A VIEWER calling the API gets 403 'You don't have permission to create pages' / 'You don't have permission to edit this page' / 'You don't have permission to delete this page'.	
Related Test Cases	US-78 : Viewer is read-only (UI and server) [Version :1]US-85 : Delete a page with sub-pages [Version :1]

Requirement: REQ-P-NOTE-08 : Network interruption

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
If a member loses connection, edits are kept locally and synchronised when the connection returns (Iteration 4 UC-S4-002 A1).	
Related Test Cases	US-86 : Offline edits sync after reconnect [Version :1]

13. Requirements Spec.: SRS-R-SHR : UC-S4-001 Course Sharing & Permissions

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (8/0)

Course owner shares a course with other students as VIEWER or CONTRIBUTOR. Source: UniShare_Iteration_4.md Use Case Diagram 1 + Use Case 1; src/server/api/routers/course.ts (inviteCollaborator, acceptInvitation, rejectInvitation, getPendingInvitations); resource.ts.

Requirement: REQ-SHR-01 : Only the course owner can invite collaborators

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)

The system shall allow only the student who created a course to invite collaborators. Any other user shall receive 'Only the owner can invite collaborators'. Inviting to a non-existent course shall return 'Course not found'.

Related Test Cases	US-3 : Non-owner cannot invite collaborators [Version :1]US-6 : Invite to a course that does not exist [Version :1]

Requirement: REQ-SHR-02 : Invalid invitations are rejected

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)

The system shall reject an invitation to the owner himself ('You can't invite yourself') and a second invitation to a user already invited to the same course ('User already invited').

Related Test Cases	US-4 : Owner cannot invite himself [Version :1]US-5 : Duplicate invitation to the same student is rejected [Version :1]

Requirement: REQ-SHR-03 : Invitation carries a permission level

Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)

The owner shall choose exactly one permission level, VIEWER or CONTRIBUTOR. A new invitation is stored with status PENDING and the invitee receives a COURSE_INVITATION notification '<owner> invited you to collaborate on <course>'.

Related Test Cases	US-1 : Owner invites same-faculty student as VIEWER [Version :1]US-2 : Owner invites same-faculty student as CONTRIBUTOR [Version :1]

Requirement: REQ-SHR-04 : Invitee accepts or declines a pending invitation

Version :	1
revision:	1

Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The invitee shall see pending invitations under Settings > Pending Invitations and can Accept or Decline. Accept sets status ACCEPTED and joinedAt and the course appears in the invitee's course list; Decline sets REJECTED. The owner receives 'Invitation accepted' / 'Invitation declined'. Only PENDING invitations can be answered.	
Related Test Cases	US-9 : Invitee accepts a pending course invitation [Version :1]US-10 : Invitee declines a pending course invitation [Version :1]US-11 : Answered invitation cannot be accepted or declined again [Version :1]US-15 : Non-collaborator and declined user cannot open a private course [Version :1]US-35 : Click invitation notification marks read and opens Pending Invitations [Version :1]

Requirement: REQ-SHR-05 : VIEWER has read-only access	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A VIEWER collaborator can open the shared course, view all resource cards and download files but cannot add, edit or delete resources or upload files.	
Related Test Cases	US-12 : VIEWER can view and download but cannot modify [Version :1]US-15 : Non-collaborator and declined user cannot open a private course [Version :1]

Requirement: REQ-SHR-06 : CONTRIBUTOR can add but not delete	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A CONTRIBUTOR can add/update resources and upload files to cards that allow uploads, but cannot delete resources ('Only the course owner can delete resources') or delete the course.	
Related Test Cases	US-13 : CONTRIBUTOR can add resources and upload files [Version :1]US-14 : CONTRIBUTOR cannot delete resources or the course [Version :1]

Requirement: REQ-SHR-07 : Owner manages collaborators	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
On Courses > <course> > Collaborators the owner can invite by email, change a collaborator's permission (VIEWER <-> CONTRIBUTOR) and remove a collaborator.	
Related Test Cases	US-1 : Owner invites same-faculty student as VIEWER [Version :1]US-2 : Owner invites same-faculty student as CONTRIBUTOR [Version :1]US-8 : Owner changes collaborator permission and removes collaborator [Version :1]

Requirement: REQ-SHR-08 : Faculty-restricted sharing	
Version :	1

revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The system shall validate that the invited student belongs to the same faculty as the owner (Iteration 4: 'System validates student is in same faculty').	
Related Test Cases	US-7 : Invitation to a student from a different faculty is blocked [Version :1]

14. Requirements Spec.: SRS-R-ART : UC-S4-003 Public Articles

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (10/0)

Students write articles that move DRAFT -> PUBLISHED -> ARCHIVED and are readable publicly. Source: UniShare_Iteration_4.md Use Case Diagram 3 + Use Case 3; src/server/api/routers/article.ts; src/app/(student)/my-articles, src/app/articles.

Requirement: REQ-ART-01 : Article input validation	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Title is mandatory, 1 to 200 characters; Excerpt optional, max 500 characters; Cover image optional and must be a valid URL; Tags optional, comma-separated. Empty title shows 'Please enter a title for your article'.	
Related Test Cases	US-17 : Title length boundaries (0, 1, 200, 201 characters) [Version :1]US-18 : Excerpt length boundaries (500 / 501 characters) [Version :1]US-19 : Cover image URL validation [Version :1]

Requirement: REQ-ART-02 : New article starts as DRAFT	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Saving a new article stores it with status DRAFT, the author as current user, an estimated read time (words/200, rounded up) and shows 'Article saved as draft'.	
Related Test Cases	US-16 : Create article saved as draft (valid data) [Version :1]

Requirement: REQ-ART-03 : Unique URL slug	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The slug is generated from the title (lower-case, punctuation removed, spaces -> '-'). If the slug exists the system appends -1, -2, ... On title change the slug is regenerated.	
Related Test Cases	US-20 : Duplicate titles get unique slugs [Version :1]US-21 : Author edits own article; tags are replaced [Version :1]

Requirement: REQ-ART-04 : Publish workflow	
Version :	1
revision:	1
Author:	admin

Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the author can publish, and only a DRAFT article can be published ('Only draft articles can be published'). Publishing sets PUBLISHED and publishedAt and shows 'Article published!'.	
Related Test Cases	US-23 : Publish a draft article [Version :1]US-24 : Only DRAFT articles can be published [Version :1]US-27 : Non-author cannot publish, archive or delete [Version :1]

Requirement: REQ-ART-05 : Archive and delete own article	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the author can archive ('Article archived') or delete ('Article deleted successfully') an article; other users get 'You can only archive/delete your own articles'.	
Related Test Cases	US-25 : Archive own published article [Version :1]US-26 : Delete own article [Version :1]US-27 : Non-author cannot publish, archive or delete [Version :1]

Requirement: REQ-ART-06 : Edit own article	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Only the author can edit an article ('You can only edit your own articles'). Saving shows 'Changes saved'. Tags supplied on edit replace the previous tags.	
Related Test Cases	US-21 : Author edits own article; tags are replaced [Version :1]US-22 : User cannot edit another user's article [Version :1]

Requirement: REQ-ART-07 : Public article listing	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The /articles page lists only PUBLISHED articles, newest publishedAt first, 12 per page by default (page size 1-50). Search matches title or excerpt case-insensitively; filter by tag; the tag list shows only tags having at least one published article.	
Related Test Cases	US-25 : Archive own published article [Version :1]US-29 : Listing shows only published, newest first, with pagination [Version :1]US-30 : Search and tag filter [Version :1]

Requirement: REQ-ART-08 : Article access control	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case

Coverage	100% (1/1)
A PUBLISHED article is readable by anyone, including visitors not logged in. DRAFT and ARCHIVED articles are readable only by the author; others get 'This article is not published'. Unknown slug gives 'Article not found'.	
Related Test Cases	US-28 : Access control by status and viewer [Version :1]

Requirement: REQ-ART-09 : View counter	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Opening a PUBLISHED article increments its view count by 1; DRAFT/ARCHIVED articles are not counted.	
Related Test Cases	US-31 : View counter increments only for published articles [Version :1]

Requirement: REQ-ART-10 : My Articles	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
My Articles lists all of the author's articles (any status), most recently updated first, filterable by DRAFT / PUBLISHED / ARCHIVED.	
Related Test Cases	US-16 : Create article saved as draft (valid data) [Version :1]US-32 : My Articles status filter [Version :1]

15. Requirements Spec.: SRS-R-NOT : UC-S4-004 Notifications

revision	1
Author	admin
Type	User Requirement Specification
Total count of requirements (Coverage)	N/A (9/0)

In-app notifications with unread badge and user preferences. Source: UniShare_Iteration_4.md Use Case Diagram 4 + Use Case 4; src/server/api/routers/notification.ts; src/components/notification-bell.tsx; src/app/(student)/settings/page.tsx.

Requirement: REQ-NOT-01 : Event notifications	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The system shall create a notification for the recipient when a course invitation is sent (COURSE_INVITATION), accepted (INVITATION_ACCEPTED) or declined (INVITATION_REJECTED).	
Related Test Cases	US-1 : Owner invites same-faculty student as VIEWER [Version :1]US-9 : Invitee accepts a pending course invitation [Version :1]US-10 : Invitee declines a pending course invitation [Version :1]

Requirement: REQ-NOT-02 : Unread badge	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
The bell icon shows the number of unread notifications; no badge at 0; values above 9 show '9+'.	
Related Test Cases	US-33 : Unread badge boundaries (0, 1, 9, 10) [Version :1]

Requirement: REQ-NOT-03 : Recent notifications dropdown	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Opening the bell lists the user's 20 most recent notifications, newest first; with none it shows 'No notifications yet'.	
Related Test Cases	US-34 : Dropdown lists 20 most recent, newest first [Version :1]

Requirement: REQ-NOT-04 : Mark one as read	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case

Coverage	100% (1/1)
Clicking a notification marks it read and decreases the badge by 1. A course/timetable invitation notification opens Settings > Pending Invitations (/settings?tab=invitations).	
Related Test Cases	US-35 : Click invitation notification marks read and opens Pending Invitations [Version :1]

Requirement: REQ-NOT-05 : Mark all as read	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
'Mark all as read' sets every unread notification of the user to read; badge disappears.	
Related Test Cases	US-36 : Mark all as read [Version :1]

Requirement: REQ-NOT-06 : Notification privacy	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
A user can only list, count and mark his own notifications; another user's notification id gives 'Notification not found'.	
Related Test Cases	US-37 : User cannot read or mark another user's notification [Version :1]

Requirement: REQ-NOT-07 : Notification preferences	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Settings > Notification Preferences lets the user switch in-app and email notifications on/off for Class reminders, Collaboration invites, Audit log alerts and System notifications. Defaults are created on first visit; saved values persist after reload.	
Related Test Cases	US-39 : Default preferences are created on first visit [Version :1]US-40 : Save in-app / email preferences per type [Version :1]

Requirement: REQ-NOT-08 : Reminder timing range	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
Class reminder timing accepts 5 to 30 minutes before class; the UI offers 5, 10, 15 and 30.	
Related Test Cases	US-41 : Class reminder timing boundaries [Version :1]

Requirement: REQ-NOT-09 : Full notification history	
Version :	1
revision:	1
Author:	admin
Status	Valid
Type	Use Case
Coverage	100% (1/1)
'View all' opens the full notification list, paginated (page size 1-100, default 20).	
Related Test Cases	US-38 : View all notifications page with paging [Version :1]